

Blacktech Full Automation Stack

Blacktech Solutions Corp — Full Automation Stack

Table of Contents

Last updated: May 25, 2026
Built on: Raspberry Pi 5 + External Hard Drive (Blacktech_Drive)

1. The Brain: Hermes Agent (Pi)

What it is: AI assistant running 24/7 on your headless Raspberry Pi
Location: /home/allenai/.hermes/profiles/derrell-black/ **What it does:** - Reads/responds to your Telegram messages instantly - Runs scheduled cron jobs (social, invoices, reminders) - Executes scripts (browser automation, email, file management) - Remembers your preferences across sessions (MEMORY.md)

2. Data Storage Layer

A) External Hard Drive — Blacktech_Drive

Mount: /media/allenai/Expansion/Blacktech_Drive/ **Purpose:** All business files live here, not on the Pi

Folder	What's Inside
1-Admin/	Invoices, estimates, reports, checklists, lead files
1-Admin/QBO_Reports/	Invoice templates, London Town schedule, Electrification estimate
1-Admin/Leads/	Contact files (e.g., Ali_Khalifa_60482.md)
1-Admin/System_Architecture/	This document
(other folders exist)	Marketing, accounting, project files

B) Pi Local Storage

Path: ~/.hermes/profiles/derrell-black/ | File | Purpose | |---|---|
| scripts/ | Python scripts (QBO checker, email bots, social
generators) | | qbo_cookies.json | Session cookies for QuickBooks
Online automation | | .env.qbo | QBO login credentials | | cache/images/
| Screenshots and images you send via Telegram |

3. Communication Layer

A) Email System (HostGator)

Server: mail.blacktechsolutionscorp.com **Password:** Newproject26\$

Address	Status	Forwards To	Aut Rep
support@blacktechsolutionscorp.com	ACTIVE	— (primary)	No
vendors@blacktechsolutionscorp.com	ACTIVE	support@	No
invoices@blacktechsolutionscorp.com	ACTIVE	support@	Yes
automation@blacktechsolutionscorp.com	ACTIVE	— (Zapier/tools)	No

How the Pi uses email: - Checks invoices@ for payment notifications
- Sends invoice reminders (e.g., Cleo, Production Down) - Sends lead
follow-ups (e.g., Ali Khalifa) - Alerts you via Telegram for real
customer emails

B) Telegram (Right Now)

Your main interface to the Pi. You text the Pi, it does the work.

C) Zapier — Facebook Auto-Post

What it is: Bridge between the Pi and Facebook **How it works:** 1. Pi
writes a social post 2. Pi sends it via curl to a Zapier webhook URL 3.
Zapier pushes it to Blacktech's Facebook Page **Status:** ⚠ Broken —
webhook returned "please unsubscribe me" — needs rebuild

D) Twilio — (NOT YET SET UP)

Status: ✗ Not configured **What it would do:** Send SMS text
messages to leads/customers directly from the Pi **What you'd need:**
Twilio account, phone number, API key

E) Cloudflare — (NOT YET SET UP)

Status: ✗ Not configured **What it would do:** - Host your website/landing page - Tunnel into your Pi from outside your home network (Cloudflare Tunnel) - Protect against spam/bots - Custom domain: blacktechsolutionscorp.com

4. Business Layer

A) QuickBooks Online (QBO)

What it is: Your accounting + invoicing system **How Pi connects:** - Browser automation with Playwright + saved cookies - Reads QBO cookie file from laptop (you export from Edge on laptop, paste into Telegram)

What the Pi does: - Searches invoices (e.g., “Does Robbins invoice exist?”) - Generates invoice templates (London Town, Electrification) - Monitors overdue invoices (daily report) - ⚠ Creating invoices is still manual on laptop (Playwright too fragile)

Manual steps for QBO cookie refresh: 1. Log in to QBO on laptop 2. Export cookies from Edge DevTools 3. Paste into Telegram 4. Pi saves to qbo_cookies.json

B) Invoice & Collections Tracking

What you’ve built: - London_Town_Invoices_Template.md — 15 invoices × \$9,200 = \$138,000 - Electrification_Rough_In_15_Units.md — 15 units × \$9,488 = \$142,313 - Invoice_Submission_Checklist.md — Drexel/Dorchester tracker

Collections workflow: - Pi runs daily check → texts you total overdue count + dollar amount - You say “mark paid” or “void” → Pi updates QBO status

5. Marketing Automation Layer

A) Daily Social Posts

Cronjob: blacktech-daily-social — runs at **8 AM every day** **What it does:** - Writes a Facebook post about electrical services - ⚠ Currently broken (Zapier webhook dead) - Normally would auto-post; for now Pi texts you the draft to copy/paste

B) Weekly Batch Content

Cronjob: blacktech-weekly-batch — runs at **7 AM Mondays** **What it does:** Generates full week of social media + Mailchimp content

C) Saturday Post

Cronjob: blacktech-social-saturday — runs at **10 AM Saturdays**
What it does: Weekend-themed electrical post

D) Mailchimp Daily Email

Cronjob: blacktech-daily-mailchimp — runs at **9 AM every day** **What it does:** Drafts a daily email for Mailchimp list

E) Monthly Newsletter

Cronjob: blacktech-monthly-newsletter — runs at **9 AM on the 1st**
What it does: Builds full monthly email

6. System Monitoring & Security

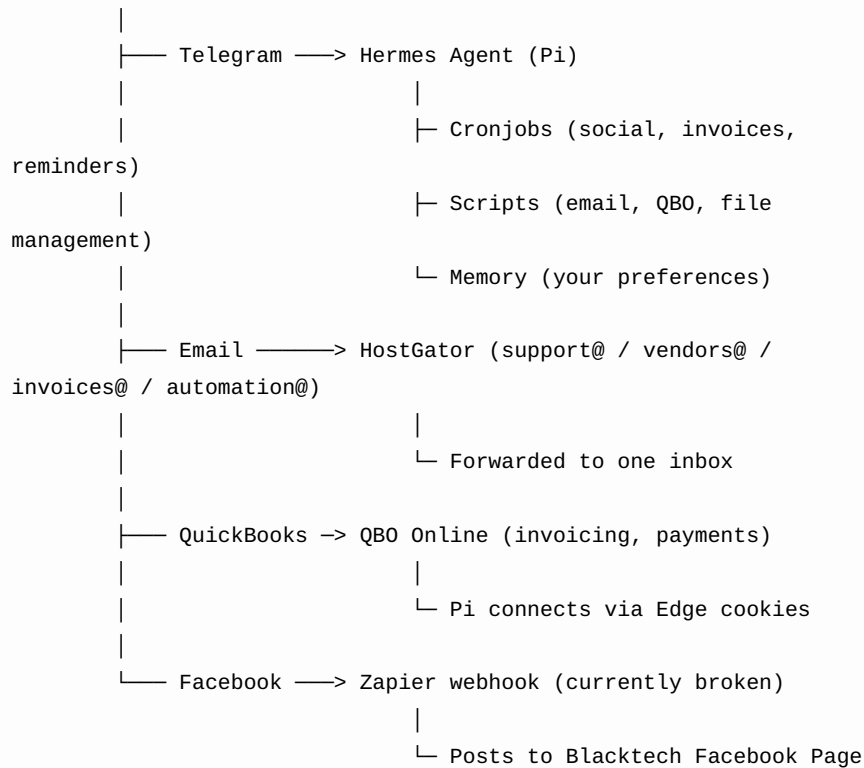
Cronjob	Schedule	What It Checks
Weekly Network Security Scan	Sundays at 4 AM	Pi security status
4AM Network Status Report	Daily at 4 AM	Pi CPU, memory, disk usage
Daily Invoice Report	Daily at 8 AM	QBO overdue invoices + text alert

7. Lead Management Workflow

Example: Ali Khalifa (LG Pro Dealer) 1. **Lead comes in** (screenshot from LG Pro Dealer) → sent to Telegram 2. **Pi reads it** → drafts email → you approve 3. **Pi sends email** from support@blacktechsolutionscorp.com 4. **Pi saves contact** to /Blacktech_Drive/1-Admin/Leads/ 5. **Pi sets reminder** — call at scheduled time tomorrow 6. **Cronjob fires** → Pi reminds you in Telegram

8. How It All Connects

YOU (Laptop / Phone)



EXTERNAL DRIVE

└ All business files, templates, leads, reports

9. What's Working vs. What's Broken

System	Status	Notes
Hermes Agent (Pi)	✓ Working	24/7, answers instantly
Telegram messaging	✓ Working	Your primary interface
Email (send/receive)	✓ Working	Sends from Pi, checks inboxes
Email forwarding	✓ Working	invoices@ + vendors@ → support@
Email auto-reply	✓ Working	invoices@ auto-replies now
QBO cookie login	✓ Working	Fresh cookies from laptop

QBO invoice search	✓ Working	Pi can find invoices
QBO create invoice	✗ Broken	Too fragile; do it manually on laptop
Zapier → Facebook	✗ Broken	Webhook dead; needs rebuild
Twilio SMS	✗ Not set up	Would cost ~\$15/month
Cloudflare	✗ Not set up	Website security + Pi tunneling
Google Drive sync	✗ Not done	Could sync external drive to cloud
Formspree	⚠ Needs upgrade	Hit free limit; needs Gold tier

10. Next Priorities

1. **Rebuild Zapier webhook** using `automation@blacktechsolutionscorp.com` ✓
2. **Formspree upgrade** to Gold for file attachments ✓
3. **Google Drive sync** for Drexel/Dorchester files
4. **Twilio setup** if you want SMS lead alerts
5. **Cloudflare tunnel** to access Pi from anywhere
6. **Website landing page** for Blacktech
7. **QBO invoice creation** via more reliable script

Document maintained by Hermes Agent — ask “Update the stack doc” to refresh.